

(別紙 1-4 説明書(英語))

Request for cooperation to

“Image enhancement of chest X-ray with Visibility and Fidelity Assessment”

The document describes the content of “Image enhancement of chest X-ray with Visibility and Fidelity Assessment” Please rest assured that you will not suffer any disadvantages even if you do not participate in this research. If you have any questions, please do not hesitate to ask the person in charge.

1. Overview of this study

■ Title of research

Image enhancement of chest X-ray with Visibility and Fidelity Assessment

■ Principal investigator Name, Affiliation and Position

- Takeshi Oishi, Institute of Industrial Science, University of Tokyo, Associate Professor

■ Investigator Name, Affiliation and Position/Grade

- Ryoichi Ishikawa, Institute of Industrial Science, University of Tokyo, Project Researcher
- Masataka Kagesawa, Institute of Industrial Science, University of Tokyo, Research Associate

■ Investigator in other institution Name, Affiliation and Position

- Takahiro Nakahashi, AirWater corporation, Counselor

■ Purpose of research

The purpose of this study is to develop an image processing method that improves the "visibility" of lesions in chest radiographs while maintaining image "fidelity". In other words, to make lesions easier to see while preserving the information in the original X-ray image.

It is hoped that this research will be useful in assisting physicians in medical diagnosis.

■ Research Method

You will be asked to view two images presented on a display in the dark and to make judgments and evaluations about the content of the images. The main purpose of this experiment is to compare our method with several image processing methods.

There will be two sessions of the experiment, and each session will last about 15 minutes. There will be a break between experiments.

2. Voluntary research cooperation and freedom of withdrawal

Whether or not to cooperate in this research is left to the free will of all collaborators. If you withdraw your consent after consent, please sign the consent withdrawal form and submit it to address at the end of this document. In addition, if you do not cooperate in the research, it will not cause any disadvantage to everyone.

If you withdraw your consent, the answers provided will be discarded and will not be used for further research. However, in the following cases, please understand that answer cannot be destroyed even if you withdraw your consent.

- In the case when an individual cannot be identified from the provided answer.
- In the case when data analysis has already been performed and your answer cannot be separated and discarded.

3. Protection of personal information

In this research, we will make every effort to protect personal information and respect privacy in order to prevent disadvantages to research participant. I will remove personal information such as name, address, or date of birth, etc. from your samples/information and put a new code to it so that we do not know who's material. We will store your personal information strictly. Analysis of anonymous answers may be provided to the joint research institute AirWater corporation, but personal information will not be provided.

4. Publication of research results

The data measured in this experiment (gender, age, and responses in the experiment) will be used for research on diagnostic aids in X-ray imaging. Research results may be published in academic publications (papers, presentations at academic conferences, etc.) as well as for the purpose of publicizing the research activities of the Oishi Laboratory.

5. Benefits and disadvantages for research participants

This research is intended to assist physicians in medical diagnosis. In addition to this, other applications are expected in terms of "making it easier to see certain things".

Undergoing the experiment in the dark may cause eye fatigue. It is also possible that the chest X-ray image may cause discomfort.. If you feel discomfort or fatigue, or if you feel sick, please inform us and we will stop the experiment at any time.

6. Handling policy of samples and information after research

The answers provided by you will be anonymized and used for research and analysis. In addition, the research results will be stored for at least 10 years after being published at the University of Tokyo Oishi Laboratory.

7. Your expense

You will not be responsible for the cost of this research. You will receive an honorarium of 1,500 yen including transportation expenses for your participation in this study.

8. Other

This research is conducted after review by the University of Tokyo Research Ethics Committee of University of Tokyo and approval by the Dean of the Institute of Industrial Science of University of Tokyo. The costs for this research are being paid from joint research expenses with AirWater corporation of the University of Tokyo. If you have any comments or questions, please feel free to contact us below.

Contact information

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